

SWS **261kWh**

C&I Energy Storage System



SW **75/100/110/125kW**

Three Phase On-grid Inverter

Model	SWS-P125E261-HC-G2
Battery Data	
Cell Type	LFP (LiFePO4)
Cell Capacity (Ah)	314
Module Nominal Energy (kWh)	52.25
Number of Packs	5
Rack Nominal Energy (kWh)	261.25
Rack Usable Energy (kWh)	261.25
Nominal Voltage (V)	832
Operating Voltage Range (V)	676 ~ 936
Max. Continuous Charge / Discharge Current (A)	188
Max. Charge / Discharge Current (A)	198.5
Max. Charge / Discharge Rate	0.5P
Depth of Discharge	90% ~ 100% (90% Recommended)
AC Output Data (On-grid)	
Nominal Output Power (kW)	125
Max. Output Power (kW)	137.5@400V AC, 130.6@380V AC
Nominal Apparent Power | Nominal Apparent Power Output to Utility Grid | Nominal Apparent Power from Utility Grid (kVA)	125
Max. Apparent Power (kVA)	137.5@400V AC, 130.6@380V AC
Max. Apparent Power Output to Utility Grid (kVA)	137.5@400V AC, 130.6@380V AC
Max. Apparent Power from Utility Grid (kVA)	137.5@400V AC, 130.6@380V AC
Nominal Output Voltage (V)	400 / 380, 3L / N / PE
Output Voltage Range (V)	340 ~ 440 / 323 ~ 418
Nominal AC Grid Frequency (Hz)	50 / 60
AC Grid Frequency Range (Hz)	47.5 ~ 52.5 / 57.5 ~ 62.5
Max. AC Output Current | Max. AC Current Output to Utility Grid | Max. AC Current from Utility Grid (A)	198.5
Nominal Output Current (A)	180.4@400V AC, 189.9@380V AC
Power Factor	~1 (0.8 lag to 0.8 lead)
Max. Total Harmonic Distortion	<3%
AC Output Data (Off-grid)	
Nominal Output Power (kW)	125
Max. Output Power (kW)	137.5@400V AC, 130.6@380V AC
Nominal Apparent Power | Nominal Output Apparent Power to Grid | Nominal Input Apparent Power from Grid (kVA)	125
Max. Apparent Power (kVA)	137.5@400V AC, 130.6@380V AC
Max. Output Apparent Power to Grid (kVA)	137.5@400V AC, 130.6@380V AC
Max. Input Apparent Power from Grid (kVA)	137.5@400V AC, 130.6@380V AC
Nominal Output Voltage (V)	400 / 380, 3L / N / PE
Output Voltage Range (V)	340 ~ 440 / 323 ~ 418
Nominal Output Frequency (Hz)	50 / 60
AC Grid Frequency Range (Hz)	47.5 ~ 52.5 / 57.5 ~ 62.5
Max. AC Output Current (A) | Max. AC Current Output to Grid (A) | Max. AC Current from Grid (A)	198.5
Nominal Output Current (A)	180.4@400V AC, 189.9@380V AC
Power Factor	~1 (0.8 lag to 0.8 lead)
Output THDI (@Linear Load)	<3%
Efficiency	
Max. PCS Efficiency	98.6%
System Efficiency	92.0%
Protection	
Battery Reverse Polarity Protection | Anti-islanding Protection | AC Overcurrent Protection | AC Short Circuit Protection	Integrated
AC Surge Protection	Type II
General Data	
Operation Temperature Range (℃)	-25 ~ +55
Derating Temperature (℃)	45
Storage Temperature (℃)	-20 ~ +45 (One Month); 0 ~ +35 (One Year)
Relative Humidity	10% ~ 95%
Max. Operating Altitude (m)	4000 (2000 derating)
Cooling Method	Pack: Liquid Cooling; PCS: Smart Fan Cooling
User Interface	LED, WLAN + APP
Communication Protocol	Modbus TCP, Modbus RTU
Weight (kg)	2580
Dimension (W × H × D mm)	1050 × 2250 × 1400
Noise Emission (dB)	≤70
Topology	Non-isolated
Ingress Protection Rating	IP54
Anti-Corrosion	C4 (C5 optional)
Configuration of Safety	Aerosol + water-based fire suppression, explosion-proof fan + explosion-proof plates (optional)
Charge / Discharge Switching Time	<60ms

Model	SW075KTL-T1-L-G2	SW100KTL-T1-G2	SW110KTL-T1-G2	SW125KTL-T1-G2
Input				
Max. Input Voltage (V)	800	1100	1100	1100
MPPT Operating Voltage Range (V)	180 ~ 800	180 ~ 1000	180 ~ 1000	180 ~ 1000
Start-up Voltage (V)	200			
Nominal Input Voltage (V)	370	600V @ 380 / 400Vac, 720V @ 480Vac		
Max. Input Current per MPPT (A)	42			
Max. Short Circuit Current per MPPT (A)	52.5			
Number of MPP Trackers	8	8	10	10
Number of Strings per MPPT	2			
Output				
Nominal Output Power (kW)	75.0	100.0	110.0	125.0
Nominal Output Apparent Power (kVA)	75.0	100.0	110.0	125.0
Max. AC Active Power (kW)	75.0	110.0	121.0	137.5
Max. AC Apparent Power (kVA)	75.0	110.0	121.0	137.5
Nominal Output Voltage (V)	127 / 220, 3L / N / PE or 3L / PE	220 / 380, 230 / 400, 277 / 480, 3L / N / PE or 3L / PE		
Output Voltage Range (V)	176 ~ 245	304 ~ 528	304 ~ 528	304 ~ 528
Nominal AC Grid Frequency (Hz)	50 / 60			
AC Grid Frequency Range (Hz)	45 ~ 55 / 55 ~ 65			
Max. Output Current (A)	196.9	167.1 @ 380V,158.8 @ 400V,132.3 @ 480V	183.4 @ 380V,174.7 @ 400V,145.5 @ 480V	199.4 @ 380V,198.5 @ 400V,165.4 @ 480V
Power Factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)			
Max. Total Harmonic Distortion	<3%			
Efficiency				
Max. Efficiency	98.8%	98.8%	98.8%	99.0%
European Efficiency	98.3%	98.4%	98.4%	98.5%
Protection				
PV String Current Monitoring	Integrated			
PV Insulation Resistance Detection	Integrated			
Residual Current Monitoring	Integrated			
PV Reverse Polarity Protection	Integrated			
Anti-islanding Protection	Integrated			
AC Overcurrent Protection	Integrated			
AC Short Circuit Protection	Integrated			
AC Overvoltage Protection	Integrated			
DC Switch	Integrated			
DC Surge Protection	Type II	Type II (Type I + II Optional)		
AC Surge Protection	Type II			
AFCI	Optional	Optional	Optional	Optional
Emergency Power Off	-	Optional	Optional	Optional
Rapid Shutdown	Optional			
Remote Shutdown	Integrated	Optional	Optional	Optional
PID Recovery	Optional			
Reactive Power Compensation at Night	-	Optional	Optional	Optional
Power Supply at Night	Optional	Optional	Optional	Optional
General Data				
Operating Temperature Range (℃)	-35 ~ +60			
Relative Humidity	0 ~ 100%			
Max. Operating Altitude (m)	4000			
Cooling Method	Smart Fan Cooling			
User Interface	LED, LCD (Optional), WLAN + APP			
Communication	RS485, WiFi + LAN	RS485, WiFi + LAN or 4G		
Communication Protocols	Modbus-RTU (SunSpec Compliant)			
Weight (kg)	88	85	88	88
Dimension (W × H × D mm)	930 × 650 × 300			
Topology	Non-isolated			
Self-consumption at Night (W)	<1	<2	<2	<2
Ingress Protection Rating	IP66			
DC Connector	MC4 (4 ~ 6mm²)			
AC Connector	OT / DT terminal (Max. 240mm²)			

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SWH 30/40/50kW

Three Phase Hybrid Inverter (HV)



SWH 50/75/80/99/100kW

Three Phase Hybrid Inverter (HV)



SWC 61.4/102.4/112.6kWh

C&I Battery System

Model	SWH030KH-T1-G2	SWH040KH-T1-G2	SWH050KH-T1-G2
Battery Input Data			
Battery Type		Li-Ion	
Nominal Battery Voltage (V)		500	
Battery Voltage Range (V)		200 ~ 800	
Start-up Voltage (V)		200	
Number of Battery Inputs		1	
Max. Continuous Charging Current (A)		100	
Max. Continuous Discharging Current (A)		100	
Max. Charging Power (kW)	33000	44000	55000
Max. Discharging Power (kW)	33000	44000	55000
PV String Input Data			
Max. Input Power (kW)	60000	60000	75000
Max. Input Voltage (V)		1000	
MPPT Operating Voltage Range (V)		160 ~ 850	
Start-up Voltage (V)		200	
Nominal Input Voltage (V)		620	
Max. Input Current per MPPT (A	42 / 32 / 42	42 / 32 / 42	42 / 32 / 42 / 32
Max. Short Circuit Current per MPPT (A)	55 / 42 / 55	55 / 42 / 55	55 / 42 / 55 / 42
Number of MPP Trackers	3	3	4
Number of Strings per MPPT		2	
AC Output Data (On-grid)			
Nominal Output Power (W)	30000	40000	50000
Nominal Apparent Power Output to Utility Grid (VA)	30000	40000	50000
Max. Apparent Power Output to Utility Grid (VA)	30000	40000	50000
Max. Apparent Power from Utility Grid (VA)	30000	40000	50000
Nominal Output Voltage (V)		380 / 400, 3L / N / PE	
Output Voltage Range (V)		176 ~ 276	
Nominal AC Grid Frequency (Hz)		50 / 60	
AC Grid Frequency Range (Hz)		45 ~ 55 / 55 ~ 65	
Max. AC Current Output to Utility Grid (A)	45.5 @ 380V 43.6 @ 400V	60.6 @ 380V 58.0 @ 400V	75.8 @ 380V 72.5 @ 400V
Max. AC Current From Utility Grid (A)	45.5 @ 380V 43.6 @ 400V	60.6 @ 380V 58.0 @ 400V	75.8 @ 380V 72.5 @ 400V
Power Factor		~1 (Adjustable from 0.8 leading to 0.8 lagging)	
Max. Total Harmonic Distortion		<3%	
AC Output Data (Back-up)*requires additional STS box			
Back-up Nominal Apparent Power (VA)	30000	40000	50000
Max. Output Apparent Power (VA)	33000 (36,000 @ 60s,45000 @ 10s)	44000 (48000 @ 60sec,60000 @ 10sec)	55000 (60000 @ 60sec,75000 @ 10sec)
Max. Output Current (A)	50.0 @ 380V 47.8 @ 400V	66.7 @ 380V 63.8 @ 400V	83.3 @ 380V 79.7 @ 400V
Nominal Output Voltage (V)		380 / 400, 3L / N / PE	
Nominal Output Frequency (Hz)		50 / 60	
Output THDv (@Linear Load)		<3%	
Efficiency			
Max. Efficiency		98.1%	
European Efficiency		97.5%	
Max. Battery to AC Efficiency		97.7%	
MPPT Efficiency		99.0%	
Protection			
Residual Current Monitoring | PV Reverse Polarity Protection | Battery Reverse Polarity Protection | Anti-islanding Protection			Integrated
AC Overcurrent Protection | AC Short Circuit Protection | AC Overvoltage Protection | DC Switch			Integrated
DC Surge Protection	Type II	Type II (Type I + II Optional)	
AC Surge Protection		Type II	
AFCI		Optional	
Remote Shutdown		Integrated	
General Data			
Operating Temperature Range (°C)		-35 ~ +60	
Relative Humidity		0 ~ 95%	
Max. Operating Altitude (m)		4000	
Cooling Method		Smart Fan Cooling	
User Interface		LED, WLAN + APP	
Communication with BMS		CAN	
Communication with Meter		RS485	
Communication with Portal		RS485, WiFi + LAN + Bluetooth, 4G + Bluetooth (Optional)	
Weight (kg)	62	62	62
Dimension (W × H × D mm)		520 × 660 × 260	
Topology		Non-isolated	
Self-consumption at Night (W)		<15	
Ingress Protection Rating		IP66	
Mounting Method		Wall Mounted	

Model	SWH050KH-T1-L-G2	SWH075KH-T1-G2	SWH080KH-T1-G2	SWH099KH-T1-G2	SWH100KH-T1-G2
Battery Side					
Battery Type			Li-Ion		
Nominal Battery Voltage (V)	400		600		
Voltage Range (V)	300 ~ 600		300 ~ 800		
Start-up Voltage (V)			300		
Number of Battery Inputs			2		
Max. Continuous Charging/Discharging Current (A)	85 × 2	85 × 2	100 × 2	110 × 2	110 × 2
Max. Charging/Discharging Power (kW)	50	75	88	99.99	110
PV Side					
Max. Input Power (kW)	100	150	160	200	200
Max. Input Voltage (V)	850	1000	1000	1000	1000
MPPT Operating Voltage Range (V)	160 ~ 700	160 ~ 700	160 ~ 950	160 ~ 950	160 ~ 950
Start-up Voltage (V)			200		
Nominal Input Voltage (V)	420	620	620	620	620
Max. MPPT Current (A)			42 × 8		
Max. MPPT Short Circuit Current (A)			55 × 8		
Number of MPPTs			8		
Number of Strings per MPPT			2		
AC Side (On-grid)					
Nominal Power (kW)	50	75	80	99.99	100
Max. Power (kW)	50	75	88	99.99	110
Nominal Apparent Power from/to Grid (kVA)	50	75	80	99.99	100
Max. Apparent Power from/to Grid (kVA)	50	75	88	99.99	110
Nominal Voltage (V)	127/220,3L/N/PE		220 / 380, 230 / 400, 240 / 415, 3L / N / PE		
Voltage Range (V) (according to local standard)	114 ~ 139		180 ~ 280		
Nominal Frequency (Hz)			50 / 60		
Frequency Range (Hz)			45 ~ 55 / 55 ~ 65		
Nominal Current from / to Grid (A)	131.3	1140@380Vac 1083@400Vac 1044@415Vac	121.6@380Vac 1155@400Vac 1113@415Vac	1520@380Vac 1444@400Vac 1392@415Vac	1520@380Vac 1444@400Vac 1392@415Vac
Max. Current from /to Grid (A)	131.3	1140@380Vac 1083@400Vac 1044@415Vac	1338@380Vac 1271@400Vac 1225@415Vac	1520@380Vac 1444@400Vac 1392@415Vac	1672@380Vac 1588@400Vac 1531@415Vac
Power Factor			0.8 leading ~ 0.8 lagging		
THDI			<3%		
Back-up Side					
Nominal Output Apparent Power (kVA)	50	75	80	99.99	100
Max. Output Apparent Power (kVA)	50	75	88	99.99	110
Peak Output Power without Grid (kW)	120% @ 60s;150% @ 10s	120% @ 60s;150% @ 10s	91@220V;87@230V	121@220V;116@230V	110% @ continuous; 120% @ 60s;150% @ 10s
Nominal Output Voltage (V)	127/220, 3L/N/PE		220 / 380, 230 / 400, 240 / 415, 3L / N / PE		
Nominal Output Frequency (Hz)			50 / 60		
Max. Output Current (A)	131.3	1140@380Vac 1083@400Vac 1044@415Vac	1338@380Vac 1271@400Vac 1225@415Vac	1520@380Vac 1444@400Vac 1392@415Vac	1672@380Vac 1588@400Vac 1531@415Vac
THDv (@Linear Load)			<3%		
On / Off-grid Switching Time			<10ms		
Generator Side					
Nominal Apparent Power (kVA)	50	75	80	99.99	100
Max. Apparent Power (kVA)	50	75	88	99.99	110
Nominal Voltage (V)	127/220, 3L/N/PE		220 / 380, 230 / 400, 240 / 415, 3L / N / PE		
Nominal Frequency (Hz)			50 / 60		
Frequency Range (Hz)			45 ~ 55 / 55 ~ 65		
Max. Current (A)	131.3	1140@380Vac 1083@400Vac 1044@415Vac	1338@380Vac 1271@400Vac 1225@415Vac	1520@380Vac 1444@400Vac 1392@415Vac	1672@380Vac 1588@400Vac 1531@415Vac
Efficiency					
Max. Efficiency	97.4%	98.1%	98.1%	98.1%	98.1%
European Efficiency	96.8%	97.7%	97.7%	97.7%	97.7%
Max. Battery to AC Efficiency	97.6%	98.2%	98.2%	98.2%	98.2%
MPPT Efficiency			99.9%		
Protection					
PV String Current Monitoring | PV Insulation Resistance Detection | Residual Current Monitoring | PV Reverse Polarity Protection | Battery Reverse Polarity Protection				Integrated	
Anti-islanding Protection | AC Overcurrent Protection | AC Short Circuit Protection | AC Overvoltage Protection | DC Switch				Integrated	
DC Surge Protection			Type II (Type I + II Optional)		
AFCI			Type II		
AC Surge Protection | Rapid Shutdown | Remote Shutdown			Optional		
General Data					
Operating Temperature Range (°C)			-35 ~ +60		
Relative Humidity			0 ~ 100%		
Max. Operating Altitude (m)			4000		
Cooling Method			Smart Fan Cooling		
User Interface			LED, LCD (Optional), WLAN + APP		
Communication with BMS			CAN		
Communication			RS485, WiFi + LAN + Bluetooth, 4G + Bluetooth (Optional)		
Communication Protocols			Modbus-RTU, Modbus-TCP		
Weight (kg)			97		
Dimension (W × H × D mm)			995 × 758 × 358		
Noise Emission (dB)			<60		
Ingress Protection Rating			IP66		
Anti-corrosion Class			C4		

Model	SWC61.4-H1-GC	SWC102.4-H1-GC	SWC112.6-H1-GC
Battery System			
Cell Type		LFP (LiFePO4)	
Cell Capacity (Ah)		100	
Rated Capacity (Ah)		200	
Pack Type / model		SW10.2-PACK-ACI-G2	
Pack Nominal Energy (kWh)		10.24	
Pack Configuration	2P96S	2P160S	2P176S
Pack Weight (kg)		<90	
Number of Packs	6	10	11
Nominal Energy (kWh)	61.4	102.4	112.6
Usable Energy (kWh)[1]	60	100	110
Nominal Voltage (V)	307.2	512.0	563.2
Operating Voltage Range (V)	275.52 ~ 346.56	459.2 ~ 577.6	505.12 ~ 635.36
Charging Operating Temperature Range (°C)		-20 ~ +55	
Discharging Operating Temperature Range (°C)		-20 ~ +55	
Max. Charge / Disharge Current (A)		180 / 220	
Max. Charge / Discharge Rate		0.9C / 1.1C	
Max. Charge / Discharge Power (kW)	55.2 / 67.5	92.1 / 112.6	101.3 / 123.9
Cycle Life		6000 (25 ± 2°C, 0.5C, 90%DOD, 70%EOL)	
Depth of Discharge		100%	
Efficiency			
Round-trip Efficiency		96%@100%DOD, 0.2C, 25 ± 2°C	
General Data			
Operating Temperature Range (°C)		-20 ~ +55	
Storage Temperature (°C)		+35 ~ +45 (<6 Months); -20 ~ +35 (<1 Year)	
Relative Humidity		0 ~ 100% (Condensationless)	
Max. Operating Altitude (m)		4000	
Cooling Method		Air Conditioner	
User Interface		LED	
Communication		CAN (RS485 Optional)	
Weight (kg)	<950	<1310	<1400
Dimension (W × H × D mm)		1055 × 2000 × 1055	
Ingress Protection Rating		IP55	
Anti-corrosion Class		C4 (C5-M Optional)	
Fire Safety Equipment		Aerosol (Pack & Cabinet Level)	
Certification			
Safety Regulation		IEC62619 / IEC63056 / IEC60730 / IEC62477 / VDE2510 / ISO13849 IEC62040 / N140 / EU 2023 / 1542 / UN38.3	
EMC		IEC / EN61000-6-1 / 2 / 3 / 4	

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